

Lizhong Hu

Website | Email | GitHub | Codeforces | LinkedIn

Profile

Software Engineering undergraduate in a Sino-American dual-degree program jointly operated by Quanzhou University of Information Engineering and Slippery Rock University. Interested in algorithms, software systems, information retrieval, and reliable intelligent systems, with experience in competitive programming, back-end development, Linux/Docker infrastructure, and live sports broadcast technical operations.

Research Interests

Intelligent Software Engineering; Programming Support and Online Judge Systems; Algorithms and Data Structures; Information Retrieval and Retrieval-Augmented Generation.

Education

Quanzhou University of Information Engineering

2023–2027 (*expected*)

Software Engineering (Sino-foreign Cooperation)

GPA: 3.10/4.00

Slippery Rock University

Information Systems, Bachelor of Science (BS)

GPA: 3.583/4.000

Four-year Sino-American 4+0 dual-degree undergraduate program conducted in China and officially recognized by the Ministry of Education of China.

Selected Projects

NoteLLM: Personal Learning QA System with Verifiable RAG

Ongoing

Undergraduate thesis

- Architected a verifiable RAG backend using FastAPI, PostgreSQL, and `pgvector` for cosine-similarity retrieval, with page-aware extraction and overlapping chunking across PDF, Markdown, and TXT files.
- Enforced hallucination guardrails via server-side citation validation against candidate chunks and source excerpt persistence; achieved 100% Recall@5 and a 97.1% citation match rate in benchmark evaluations.

DOMjudge & Privacy-Oriented Network Infrastructure

Sep 2024–Present

- Maintain a containerized DOMjudge for university contests; built a Cloudflare Worker subscription generator with ISP-aware routing and a dual-stack egress architecture routing IPv6 via WARP alongside native IPv4, with automated deployment and CDN speed testing.

ETi+ Smart Community Digital Governance Platform

Mar 2025–Apr 2025

- Engineered a hybrid persistence backend using PostgreSQL for landlord credit scoring and anomaly auditing, alongside MongoDB for high-throughput WeChat chat stream logging.
- Developed Flask REST APIs automating WeChat community moderation, regex nickname compliance checks, batch data ingestion, and automated Postman regression testing suites.

Technical Experience

Sinotech Xinrui (Beijing) Co., Ltd.

7–22 Sep 2026

TOC Operator / Gifu, Japan

Project: 20th Asian Games Aichi-Nagoya 2026 — broadcast technical support.

Football, Gifu Nagaragawa Stadium.

Host broadcaster: China Media Group-International Media Port (CMG-IMP).

- Checked equipment power, intercom quality, and primary/backup broadcast feeds before matches in coordination with the broadcast team.
- Monitored TOC signals during matches and provided on-site technical support throughout a 16-day assignment.
- Helped engineers isolate an optical-transmitter fault through receiver replacement and attenuation tests; the monitor feed was restored after post-match replacement and end-to-end verification.

NOI-Pre Problem Bank — Matiji

Jul 2024–Oct 2024

Problem Tester

In collaboration with China Computer Federation (CCF).

- Tested NOI-Pre programming problems; created test cases, checked solution correctness through targeted submissions, and reported issues with test data and judge results.

Competitive Programming & Awards

- 2026 National Invitational of CCPC (Fujian), The 13th Fujian Collegiate Programming Contest** — First accepted solution to Problem A.
- The 2024 ICPC Asia Kunming Regional Contest** — Onsite Contestant, 2024.
- 15th Lanqiao Cup** — First Prize, Fujian Provincial Round, Python Programming (University Group B), 2024.

Technical Skills

Languages & Algorithms: C++, Python, Java, JavaScript, Go; algorithms and data structures, competitive programming.

Back-End & Data: FastAPI, Flask, Ruoyi, PostgreSQL, MongoDB, MySQL, REST API design and integration.

Systems & DevOps: Linux, Docker, Git, SQL tooling, DOMjudge deployment, Cloudflare Workers/CDN/DNS, WARP egress routing.

Information Retrieval & AI Systems: Retrieval-Augmented Generation (RAG), LLM API integration, prompt design.